Project Report: Solar Street Light Maintenance Drive for a Rural School

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Aim

To inspect, clean and repair the 10 solar street lights around the Government Primary School, rewire faulty battery banks, record before and after readings and hand over a maintenance logbook and safety checklist to the school.

Materials and Tools

Digital multimeter (DC 0-600 V, 10 A), clamp meter, insulated screwdriver set, wire stripper, crimping tool.

4 sq mm copper battery cable (6 m), ring lugs (20), heat-shrink sleeves, petroleum jelly for terminals.

Soft brush, bucket, clean water, microfibre cloth, 6 ft fibreglass ladder.

PPE: insulated gloves, safety goggles, safety shoes, helmet.

Procedure

1. Made a list of all 10 poles (P1-P10) and noted which lights did not glow at night (P2, P5, P7, P9).
2. Switched off each light at the charge controller and covered the panel before touching any wiring.
3. Cleaned every panel with water and a soft brush in the early morning when the panels were cool.
4. Measured open-circuit voltage (Voc) and short-circuit current (Isc) of each panel at noon.
5. Measured battery voltage at rest. Found corroded terminals on P2 and P7 and a broken lug on P5.
6. Cut back the corroded cable, fitted new ring lugs, tightened terminals and applied petroleum jelly.
7. Replaced the cracked battery-to-controller cable on P9 with new 4 sq mm cable.
8. Checked the dusk sensor and the LED driver on each pole, then switched on and tested at 7 pm.

Safety Precautions

Panel covered and controller switched off before working on battery wiring; battery negative removed first.

Insulated gloves and goggles worn when handling battery terminals (acid and short-circuit risk).

Ladder footed by a second trainee at all times; no work at height when wind was strong.

No metal watch or ring; tools kept insulated; a fire extinguisher kept near the battery box.

Observations and Results

Average panel Voc before cleaning: 19.1 V; after cleaning: 21.4 V (rated 22.0 V).

Average panel Isc before cleaning: 2.1 A; after cleaning: 2.8 A (about 33% more current).

Battery rest voltage on P2 rose from 11.2 V to 12.6 V after one sunny day of charging.

Lights working: before 6 of 10, after 10 of 10. All 10 lights stayed on for at least 10 hours on the test night.

Conclusion

Dust and corroded battery terminals were the main reasons the lights failed. After cleaning and rewiring, all 10 lights work and panel current improved by about one third. I learned to read panel and battery values properly and to isolate the system before working. Monthly cleaning is needed in the dry season.

Logbook and Bill of Materials

Logbook: 18 hours over 3 days (day 1 survey and cleaning 6 h, day 2 rewiring 7 h, day 3 testing and handover 5 h).

BOM: 4 sq mm cable 6 m Rs. 420, ring lugs 20 nos Rs. 160, heat-shrink Rs. 80, petroleum jelly Rs. 60. Total Rs. 720.

A maintenance logbook and a safety checklist were handed over to the school headmaster.